

can manage your CryptDrive data by creating folders and organising them the way you want. Every document has a version history as well; in case you made an error while editing, you can retrieve the previous version to get your data back.

CryptPad has a strict privacy policy. The service calls itself a 'Zero knowledge cloud' to showcase its privacy policy. It does not collect your password and the content stored in your pads. Your data is yours alone because of the client-side encryption. Though it collects your IP address (for security reasons and to identify you as the anonymous owner of your pad), CryptPad claims it does not log the IP address to track users or usage.

CryptPad also supports self-hosting if you want to install it on your own server and be in full control over the server side as well.

EtherCalc Web Spreadsheet

This is a free software replacement for Google Sheets and can be installed on-premise. EtherCalc Web Spreadsheet has many features though not as many as a desktop application like LibreOffice Calc. You can create a Web spreadsheet and your data is saved online. Collaborative editing is possible, and the sheet can be shared after creation and adding the necessary data. Visit <https://ethercalc.org> and click on 'Create Spreadsheet' to create your spreadsheet for collaborative editing. EtherCalc uses randomly generated URLs, e.g., <https://ethercalc.org/hj6zgvux42wo> which will be your unique Web spreadsheet. You can use formats, and add the necessary mathematical, statistical and other functions that a spreadsheet needs. EtherCalc is centralised and is a Nodejs application that can be installed with the following command:

```
npm install ethercalc
./node_modules/ethercalc/bin/ethercalc
```



Figure 1: Secure WFH apps

If a Redis service is running on the same host, EtherCalc will prefer it over file based storage to improve performance. EtherCalc is also available on the Sandstrom platform.

Matrix: The privacy friendly replacement to WhatsApp, Telegram and Signal

Services like WhatsApp are proprietary and several cases of data leaks and privacy issues have been reported. So is the case with Telegram since its server side is not open source. And in the case of Signal, its centralised design is a minus point for privacy-conscious users.

Matrix is an open source project that publishes the Matrix open standard for secure, decentralised, real-time communication. This service is maintained by the non-profit Matrix.org Foundation, with the aim to create an open platform for communication.

As of June 2019, the beta version of Matrix was out, and the protocol is fully suitable for production usage. We can host our own matrix server by installing Synapse on-premise. Matrix owes its name to its ability to bridge existing platforms into a global open matrix of communication. Bridges are core to Matrix and it is designed to be as easy to write as possible, providing the highest common denominator language

to link the networks together.

The core Matrix team maintains bridges to Slack, IRC, XMPP and Gitter, while the wider Matrix community provides bridges for Telegram, Discord, WhatsApp, Facebook, Hangouts, Signal and many more.

There are various clients available for using Matrix on the Web, the desktop and various mobile platforms. The most common among these are RiotX, Riot-web/Riot-desktop, Riot.im Mobile. Riot is recently renamed as Element. See <https://element.io/>. Other popular matrix clients are Fractal for GNOME, and Quarternion. There are also various desktop clients available for terminal geeks; one among these is WeeChat.

Jitsi Meet: A free software replacement for Hangouts, Zoom and Skype

Video conferencing is an essential part of working from home, as well as for remote calling even during normal work routines. For privacy conscious users, thankfully, there is a set of tools under the JitsiProject, of which Jitsi Meet can be used for video conferencing without fear of loss of data. Jitsi Meet is an open source video conferencing solution based on WebRTC, the open standard for Web communication. Jitsi